

SignalDesk — a news ranking engine that shows its work

For anyone who needs to get across a market quickly, from several angles at once: twelve crypto and macro feeds, ranked by an auditable eight-factor score, written up and pushed daily. Polymer Tech Expo 2026.

PROBLEM STATEMENT

Twelve newsfeeds produce a few hundred crypto and macro headlines a day — mostly listicles, price-prediction bait, and the same story told six times by six outlets. The signal is in there; the cost of finding it is forty minutes every morning.

The obvious fix — hand it to a language model and ask for a summary — fails in a specific way. **The model’s selection is unauditable.** When it leads with the wrong story you cannot find out why, cannot reproduce it, and have nothing to correct — you have traded forty minutes of reading for a black box.

SOLUTION OVERVIEW

Split the job at its natural seam. **An algorithm decides what matters** — deterministic, inspectable, unit-tested. **A model writes it up**, receiving the already-ranked list and told in its system prompt that selection is not its job: it cannot re-order, add stories, or introduce a number absent from the input. Delivery today is a Telegram bot — a digest at 08:30, an immediate push for anything scoring ≥ 0.72 , three live commands — and the same engine is the build target for a **free subscription site**, where the list itself becomes the distribution channel.

- 0 • **hard filters**
clickbait, price ticks, macro noise — binary, not a penalty
- 1 • **dedupe**
TF-IDF $\cup$ rare-entity overlap, grown to transitive closure
- 2 • **eight-factor score**
keyword/recency/source $\cdot 185$ each, topicality $\cdot 130$, numeric $\cdot 100$, analysis $\cdot 100$, corroboration $\cdot 075$, asset $\cdot 040$
- 3 • **reader filters** • **adaptive top-N**
hard subject filters, then a quality gate and a per-subject cap
- 4 • **LLM write-up** → **push**
08:30 digest, ≥ 0.72 alerts, /top /digest /weights

/why 2 – "Curve exploited for \$62m"			
factor	raw	wt	adds
keyword	1.00	0.185	0.185
source_quality	0.95	0.185	0.176
recency	0.94	0.185	0.173
topicality	1.00	0.130	0.130
numeric	0.35	0.100	0.035
asset	0.50	0.040	0.020
source_count	0.00	0.075	0.000
analysis	0.00	0.100	0.000
TOTAL			0.719

Because every score carries its breakdown, /why n answers “why is this at the top” with arithmetic rather than a model’s self-explanation. A wrong lead story traces to a factor weight — visible, testable, fixable — instead of to a sampling temperature.

USE OF AI

In the product: Moonshot Kimi K3 with xAI Grok 4.3 as fallback, via TokenRouter’s OpenAI-compatible endpoint — cross-vendor on purpose, since two models from one provider are a retry, not a fallback. The system prompt forbids any claim not grounded in the supplied titles and any number absent from the input. When every provider fails, the bot emits template output from the top-ranked headlines verbatim: degraded, honest, still correctly ordered. **Building it:** Claude Code, as a pair programmer — my input was the structure (the stages, why a weight is low, which test must pass), and the code was written against that.

IMPACT & VALUE

Forty minutes of triage becomes a thirty-second read, and — unlike a summariser — the output can be challenged. Marginal cost is **one LLM call per digest**: the feeds, the Binance ticker and the Fear & Greed index are free and unauthenticated, so the data layer costs nothing to scale — which is what makes a free subscription tier viable rather than aspirational. **148 offline tests** cover the ranking path, with no network and no API key.

REFLECTIONS

Making the score explain itself changed how I built it. /why began as a nice-to-have and became the debugging tool — **every real bug here was found by reading the factor table beside a ranking, not by a test.** A `\b(exploit)\b` regex could not match “exploited”, so the \$62m hack above scored as untiered noise and ranked 7th instead of 2nd.

The instructive one was self-inflicted: the fix for a missed duplicate *over*-merged on its first live run, because ETF passes a capitalisation test and read as a rare proper noun. The repair was a rarity threshold that **scales with batch size** — three mentions in eight headlines is a third of the feed; three in forty-one is genuinely rare. Both bugs were invisible in aggregate and obvious once the intermediate state was on screen: the argument for the whole design. **Next:** the weights are reasoned, not fitted; logging rankings against what readers actually open would make them measured.